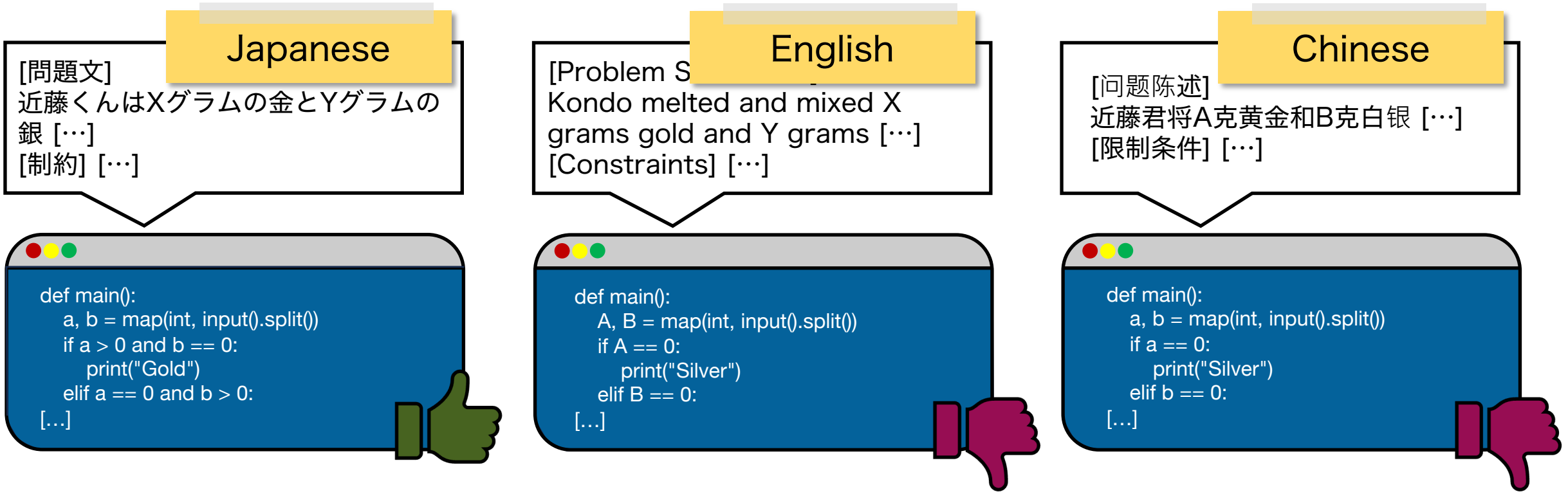


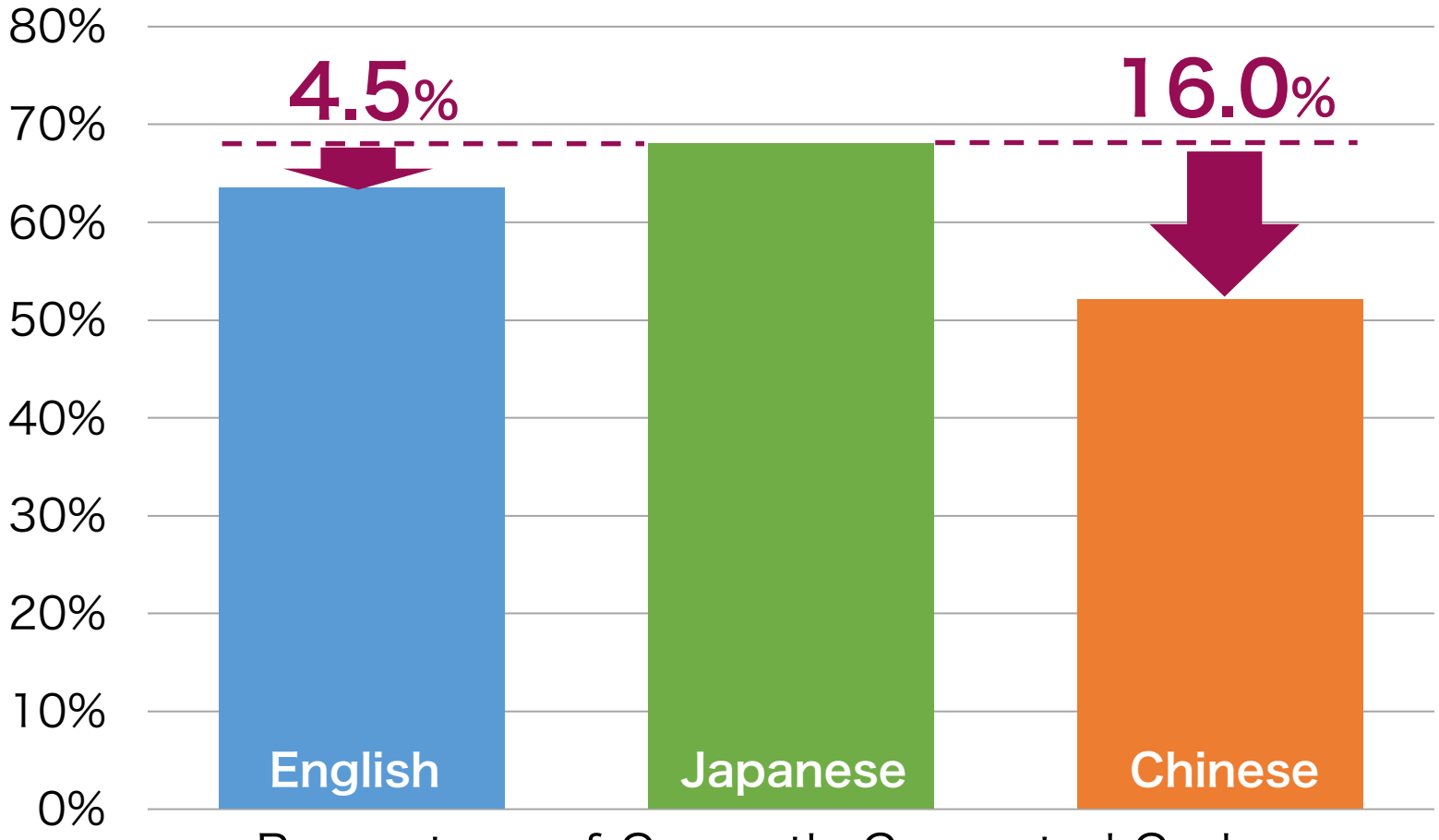
Exploring the Effect of Multiple Natural Languages on Code Suggestion Using GitHub Copilot

◎Kei Koyanagi, Dong Wang, Kotaro Noguchi, Masanari Kondo
Alexander Serebrenik, Yasutaka Kamei, Naoyasu Ubayashi

九州大学 KYUSHU UNIVERSITY
TU/e EINDHOVEN UNIVERSITY OF TECHNOLOGY

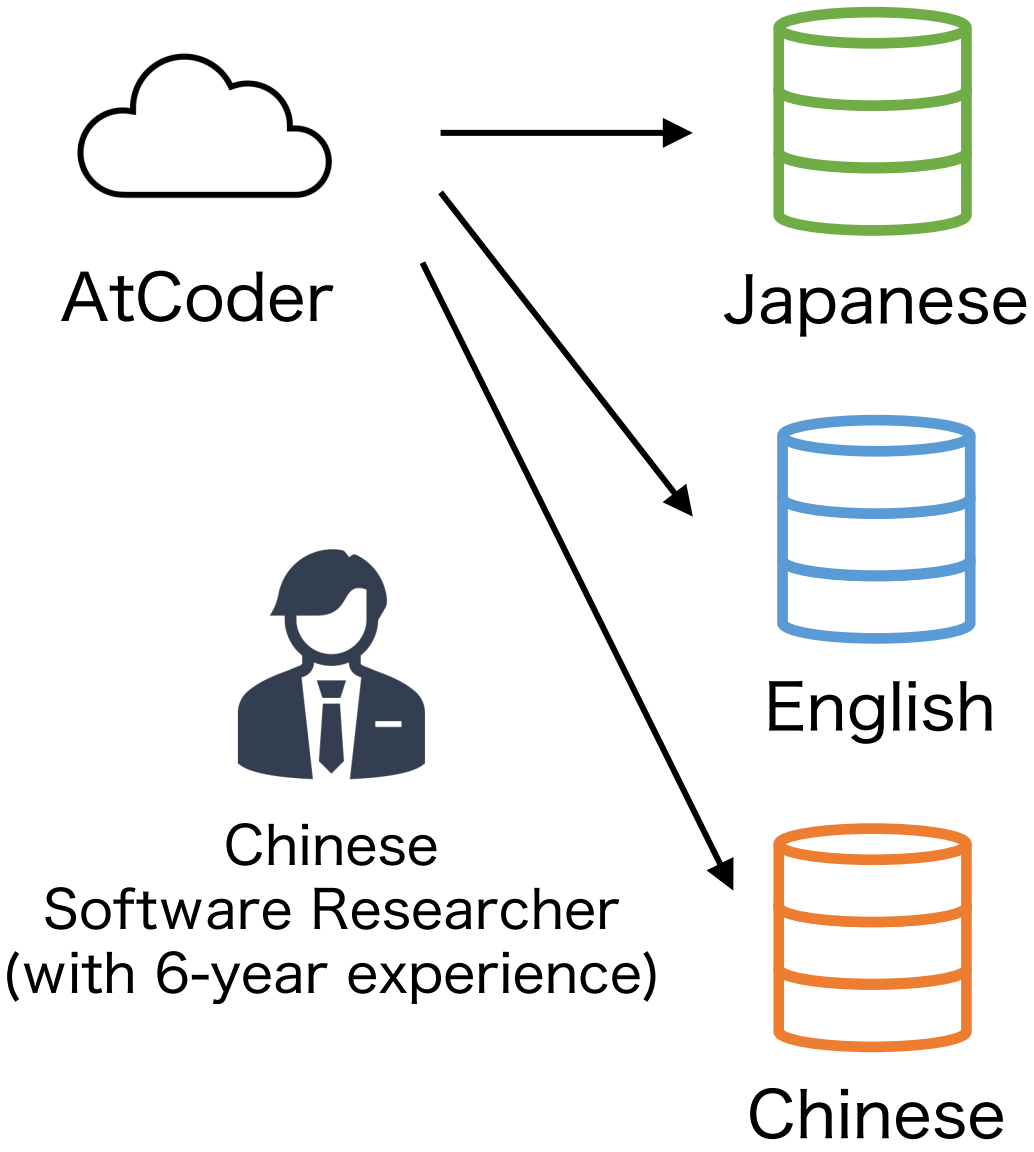
Message!

Chinese speakers may be at a disadvantage compared to English or Japanese speakers

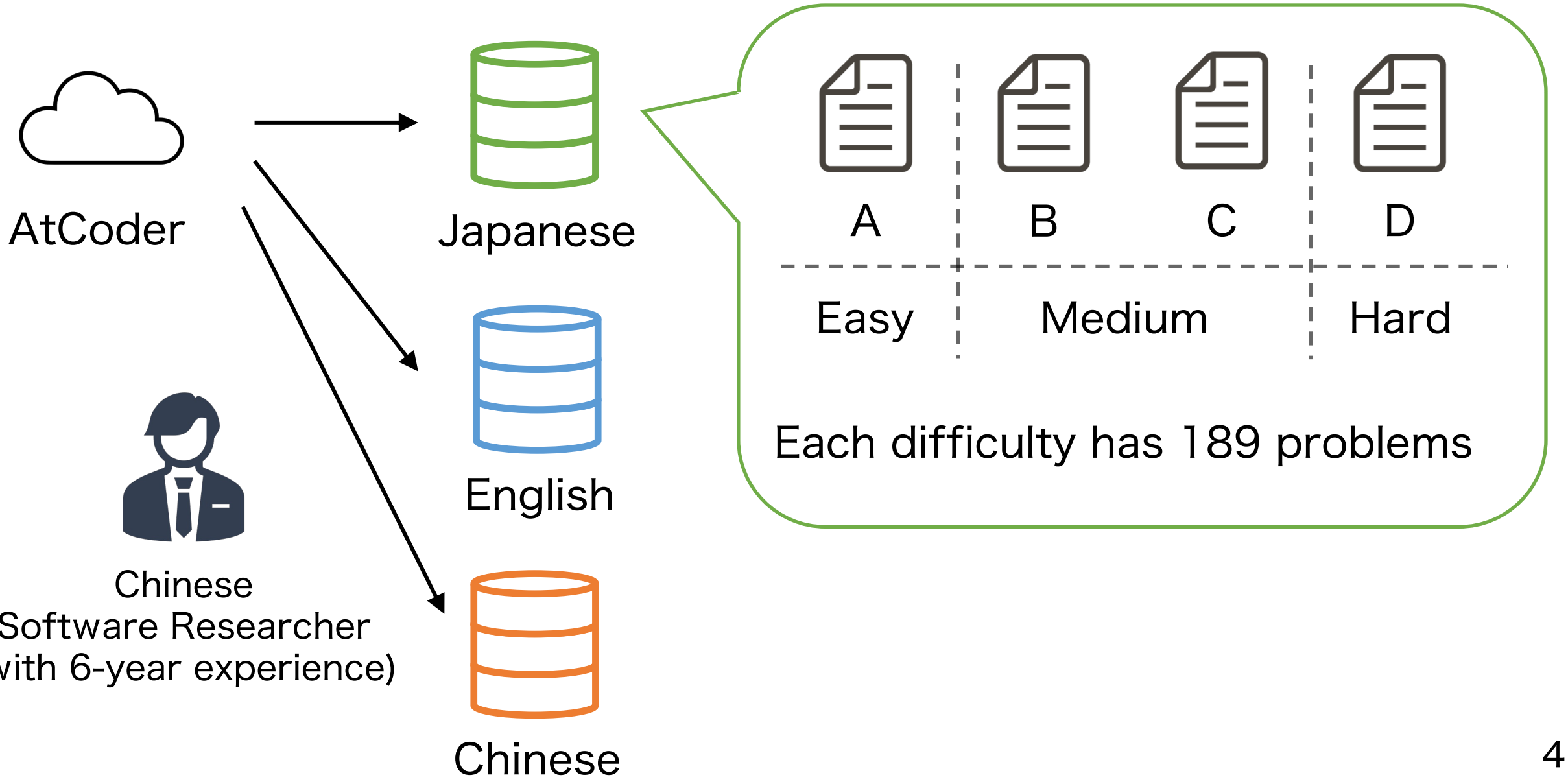


Percentage of Correctly Generated Code
in a Programming Contest

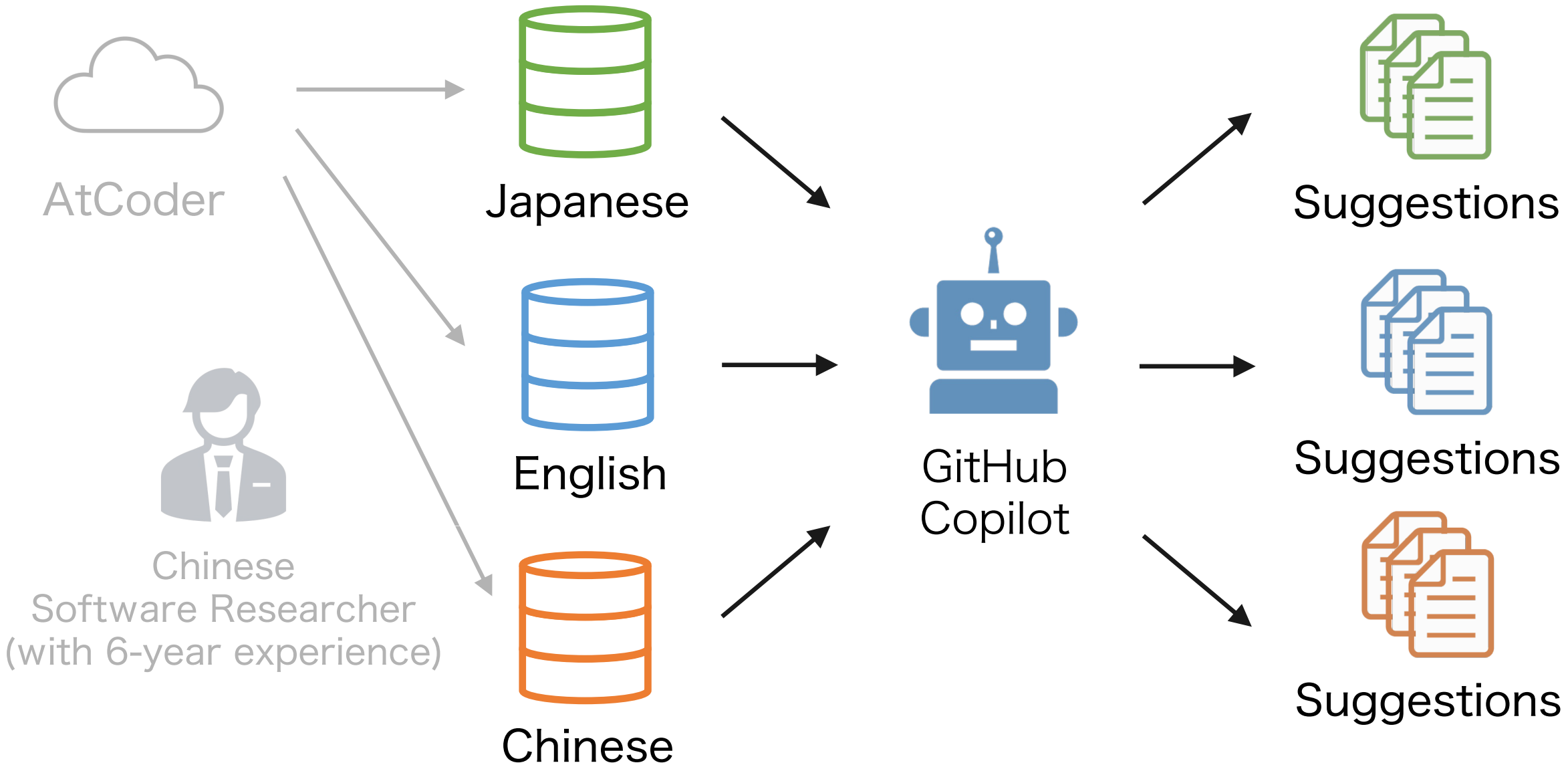
Experiment


AtCoder
Japanese
English
Chinese
Chinese
Software Researcher
(with 6-year experience)

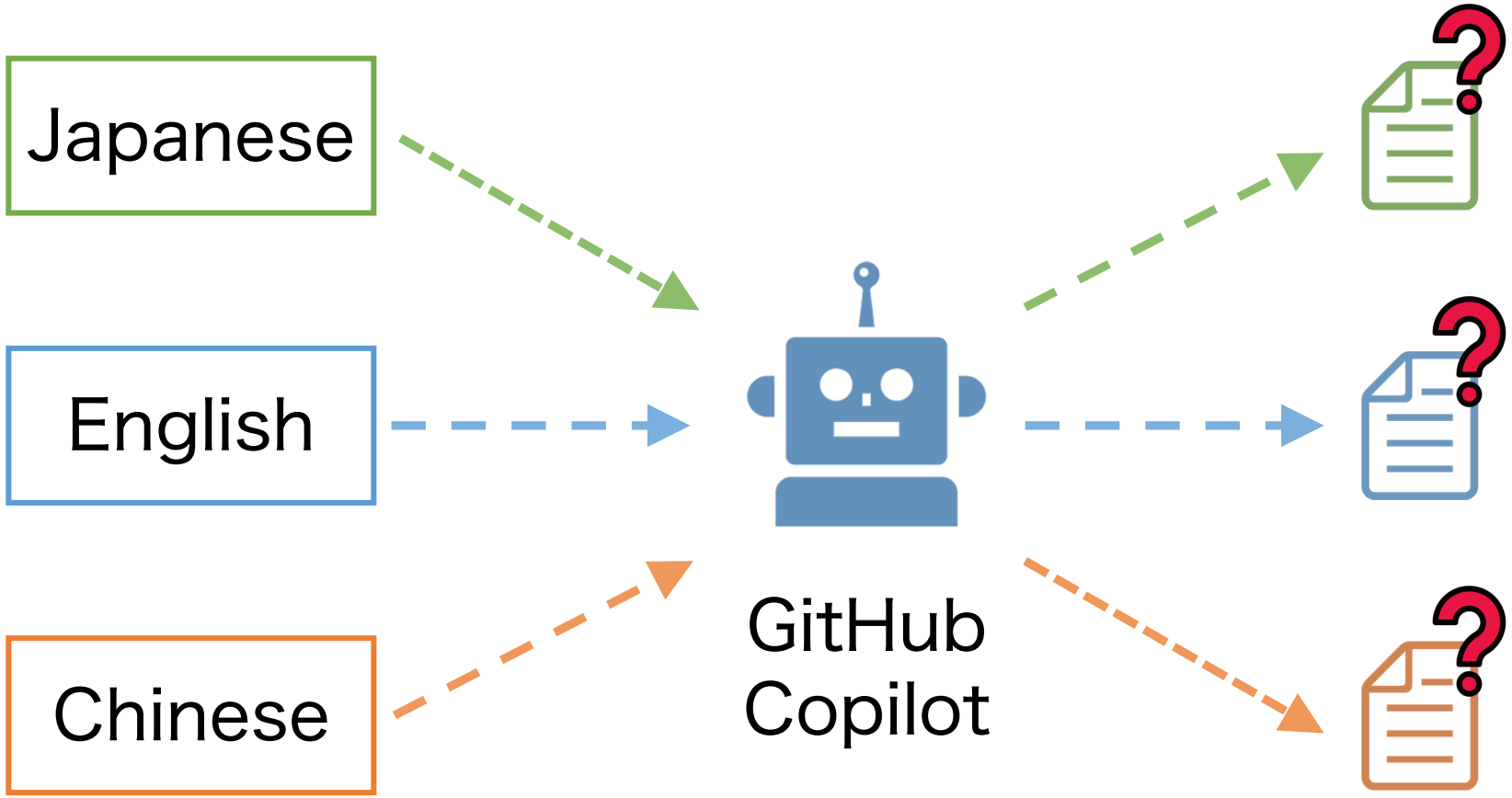
Experiment


AtCoder
Japanese
English
Chinese
Chinese
Software Researcher
(with 6-year experience)
A
B
C
D
Easy
Medium
Hard
Each difficulty has 189 problems

Experiment


AtCoder
Chinese
Software Researcher
(with 6-year experience)
Japanese
English
Chinese
GitHub
Copilot
Suggestions
Suggestions
Suggestions

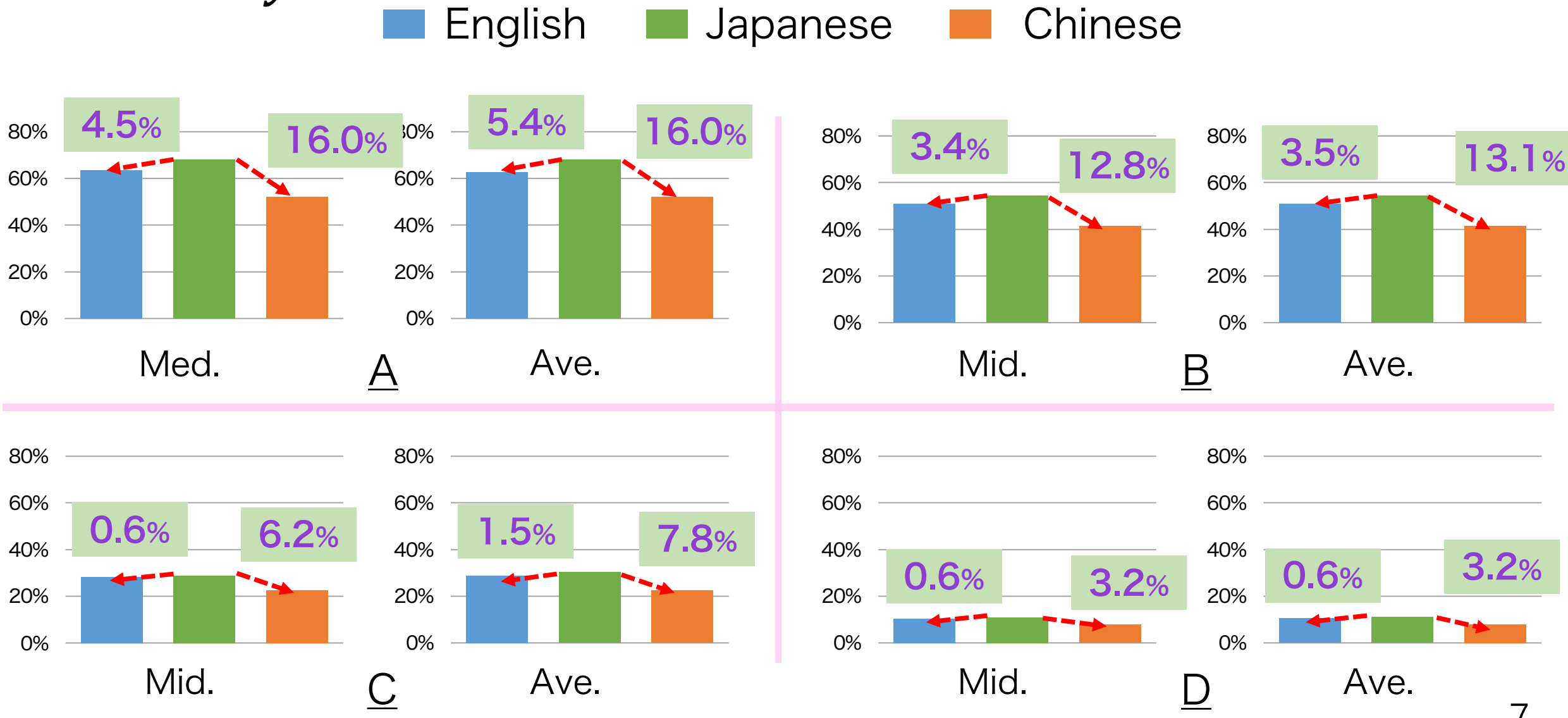
Goal: Understand how language differences affect GitHub Copilot performance



Which language is the best?

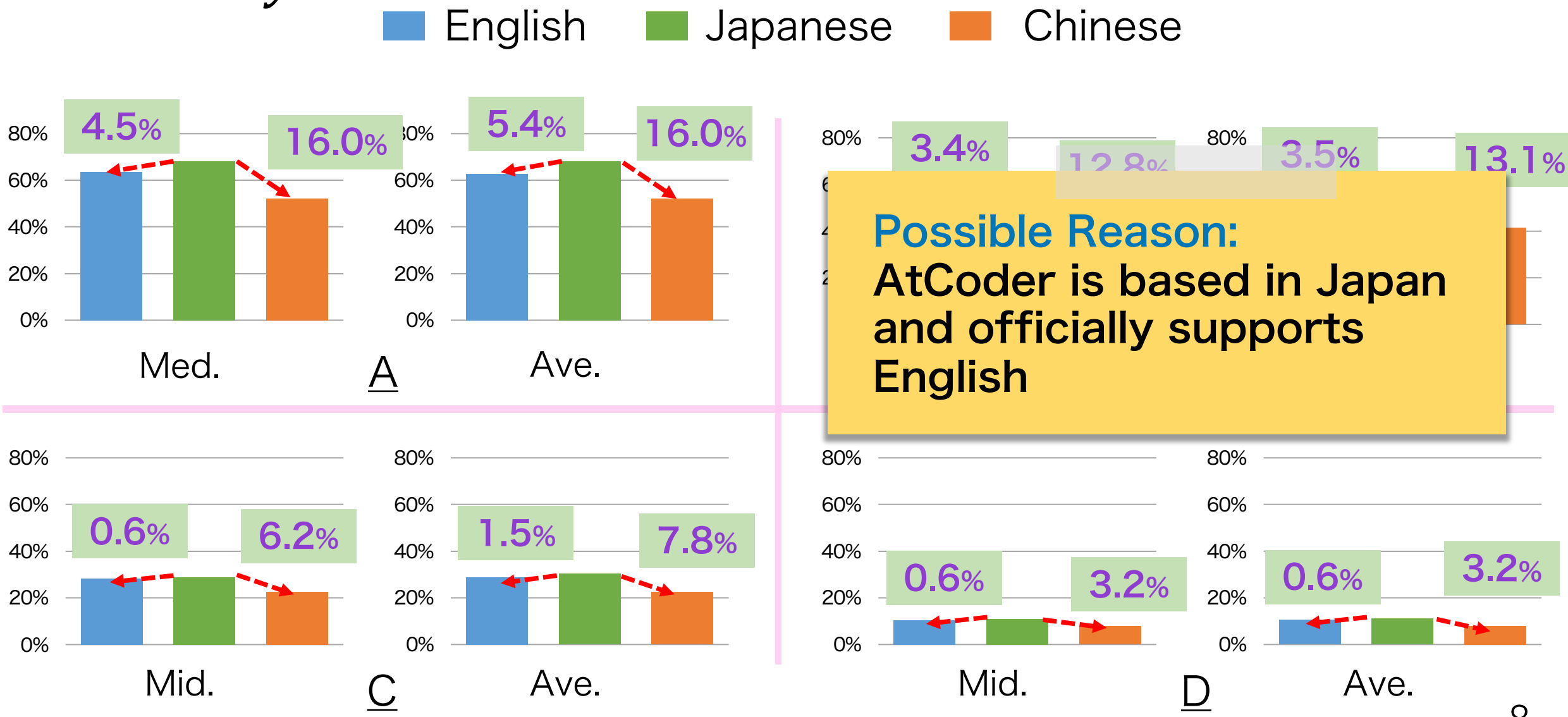
Results: Japanese recorded the highest

Accuracy



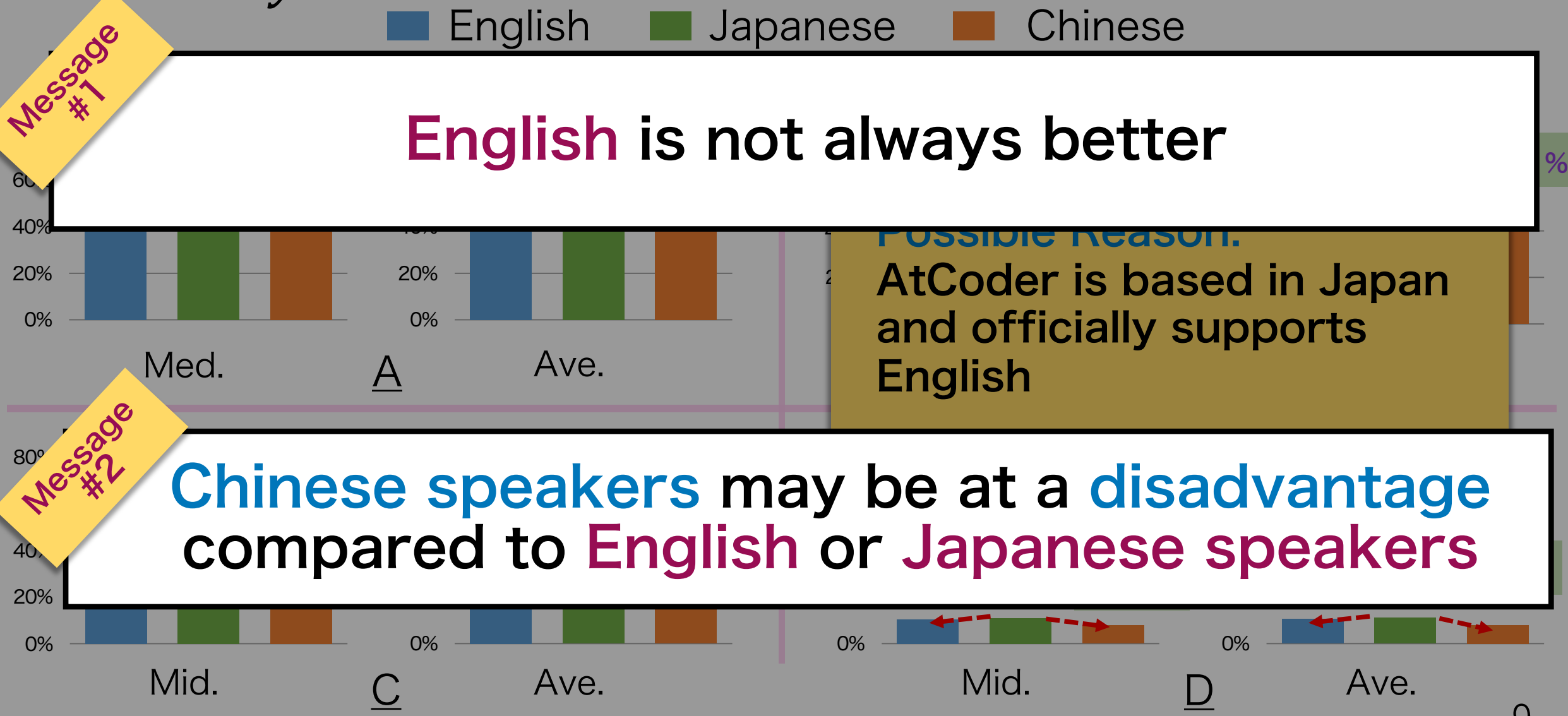
Results: Japanese recorded the highest

Accuracy



Possible Reason:
AtCoder is based in Japan and officially supports English

Results: Japanese recorded the highest *Accuracy*



Message #1

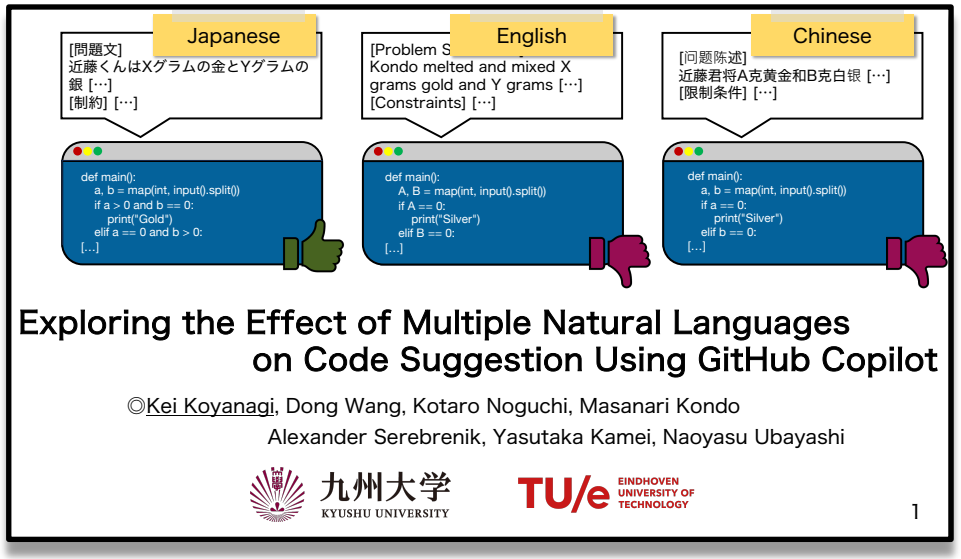
English is not always better

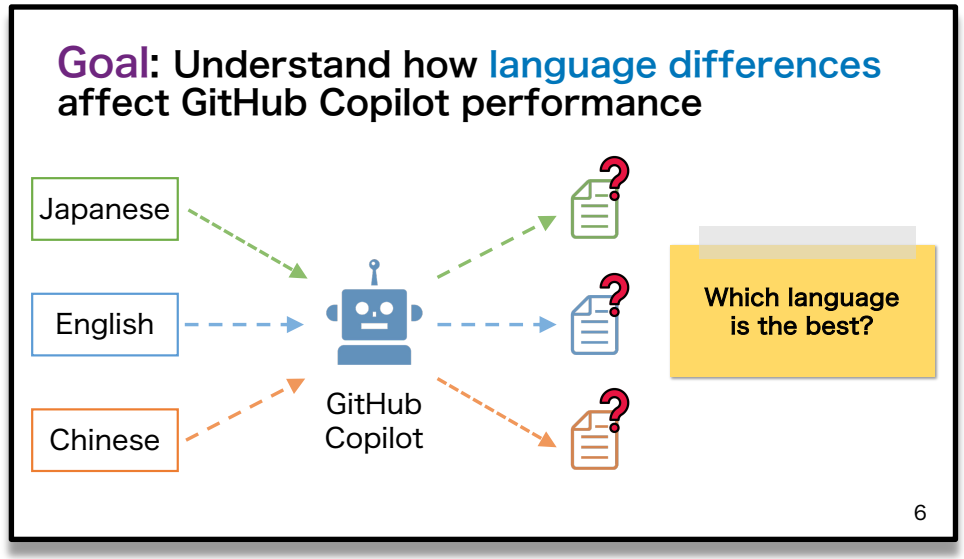
Possible Reason:
AtCoder is based in Japan and officially supports English

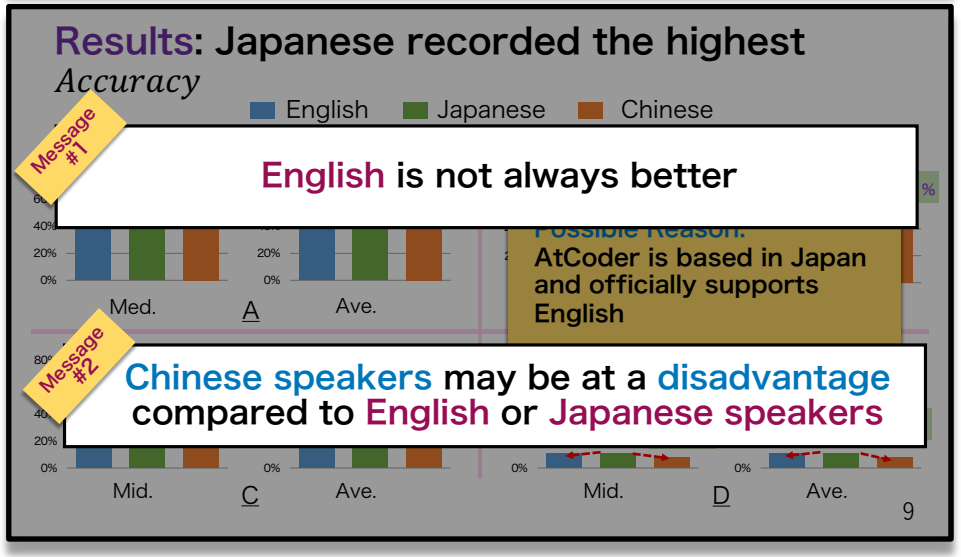
Message #2

Chinese speakers may be at a **disadvantage** compared to **English** or **Japanese speakers**

Summary: The capability varies across natural languages.







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Twitter: @masanari_kondo